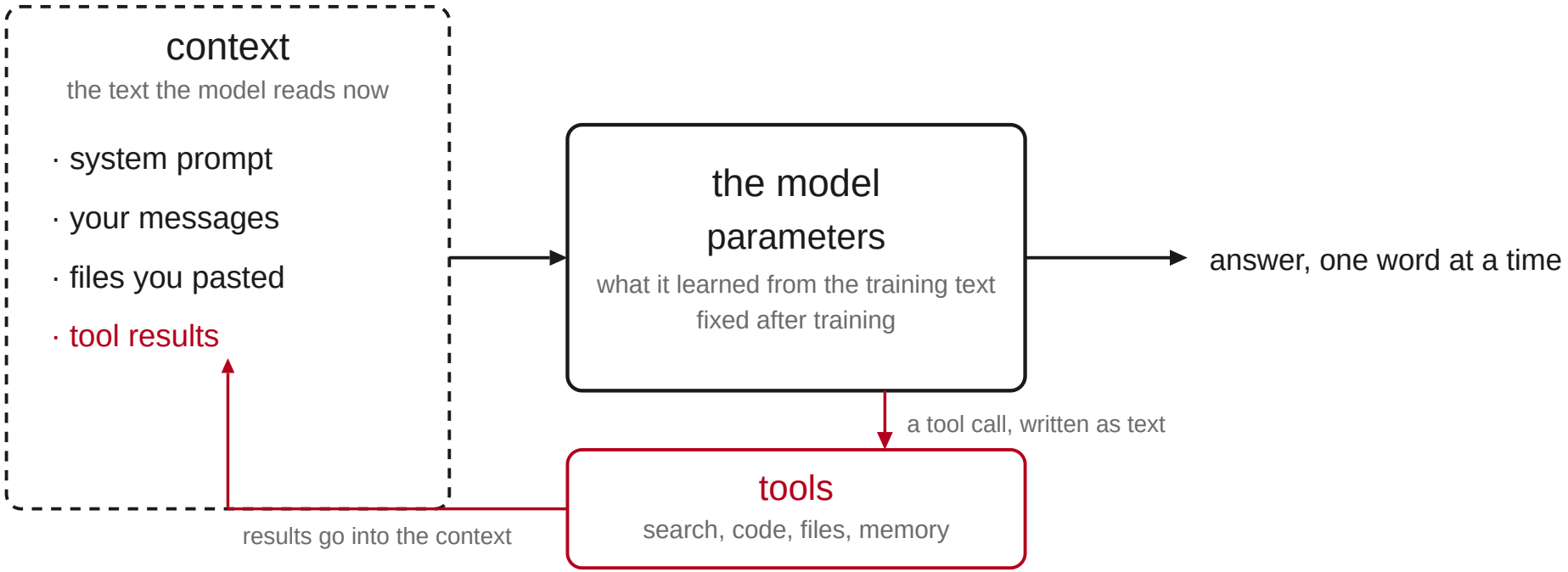
Data & AI

16. Tokens and What to Expect from a Language Model

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Last Time



- An answer comes from the parameters, the context, or a tool
- Reasoning models write many tokens before the answer

Today

- Tokens: the pieces a language model reads and writes
- Which work a model does well, which it does badly, and why

Tokens

Tokens in GPT-2

Byte pair encoding: 50,257 pieces

`In` `␣a` `␣shocking` `␣finding` `,` `␣scientist` `␣discovered` `␣a` `␣herd` `␣of` `␣unic` `orns`

- Start from the 256 values of a byte, and keep merging the pair that appears most often
- 50,000 merges: frequent words become one piece, rare words several pieces

Radford et al. (2019), Section 2.2 · pieces from the GPT-2 tokenizer (tiktoken "gpt2")

Merges in GPT-2

The first and the last entries added to the list

The first 10 merges

`␣t` `␣a` `he` `in` `re` `on` `␣the` `er` `␣s` `at`

The last 8 of the 50,000 merges

`…."` `Compar` `␣amplification` `ominated` `␣regress` `␣Collider` `␣informants` `␣gazed`

- Early merges are letter pairs that appear everywhere in English
- The list is made from counts of text, with no idea of what a word means

tiktoken encoding "gpt2", entries 256 to 265 and 50248 to 50255

A Username and Spam Words in the Token List

Rumbelow and Watkins (2023), SolidGoldMagikarp: GPT-3 at temperature 0

> Please can you repeat back the string ' SolidGoldMagikarp' to me?
>
> distribute
>
> Please can you repeat back the string ' TheNitromeFan' to me?
>
> 182

- SolidGoldMagikarp: a Reddit username, one token in GPT-2 and GPT-3
- GPT-4o tokenizer (2024): Korean spam words for escort services and gambling

출장안마 출장샵 바카라 one token each: numbers 61584, 167380, 98962 · "␣감사합니다" is 169203

What the Model Receives

The sentence from Lecture 13, as GPT-2 receives it

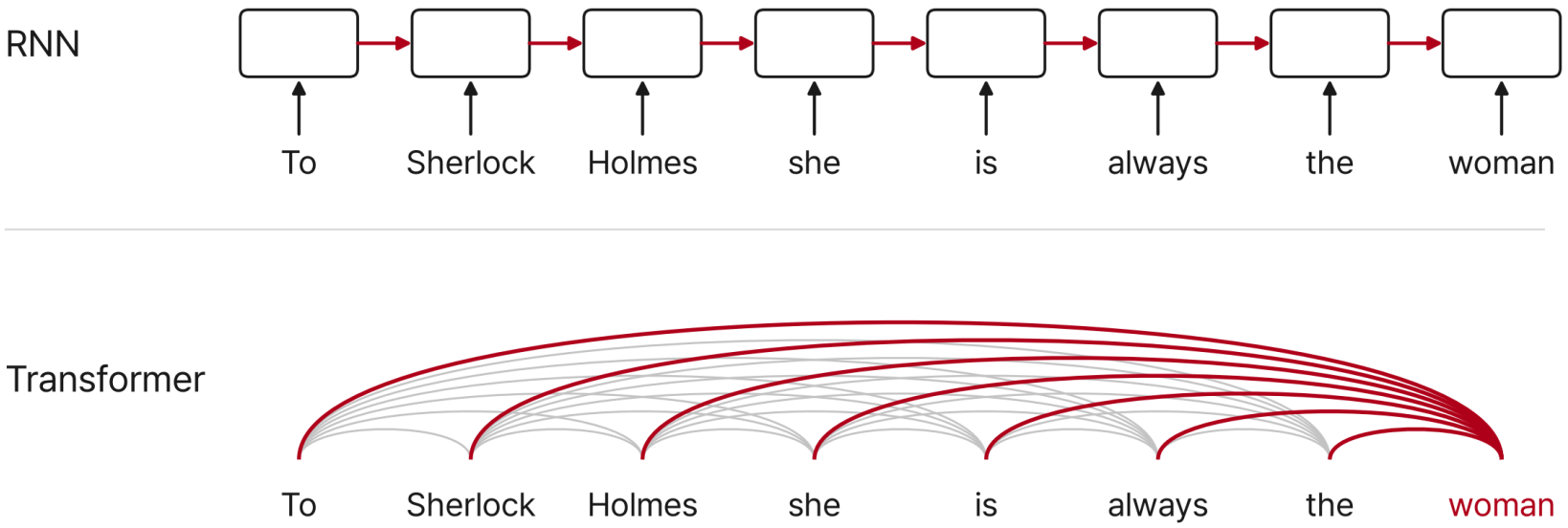
`To` `␣Sherlock` `␣Holmes` `␣she` `␣is` `␣always` `␣the` `␣woman` `.`

`2514, 25730, 17628, 673, 318, 1464, 262, 2415, 13`

- Input: 9 numbers, each one index in the list
- Output: 50,257 probabilities, one for each piece in the list
- The names RNN worked in the same way with 27 tokens

Transformer

Vaswani et al. (2017), Attention Is All You Need



- RNN: one vector passed from token to token, one step at a time
- Transformer: each token looks at every token before it (attention)
- No step waits for the one before, so a GPU computes all tokens at once

strawberry in Tokens

The GPT-4o tokenizer: one word, written in four ways

text	pieces	numbers the model receives
" strawberry" (inside a sentence)	`␣strawberry`	`101830`
"strawberry"	`st` `raw` `berry`	`302, 1618, 19772`
"Strawberry"	`Str` `aw` `berry`	`3504, 1134, 19772`
"STRAWBERRY"	`ST` `RAW` `B` `ERRY`	`1117, 46176, 33, 132354`

- The number 101830 does not say which letters are in the word

서강대학교 in Three Tokenizers

The tokenizers of GPT-2 (2019), GPT-4 (2023) and GPT-4o (2024)

GPT-2	`ec` `84` `9c` `ea` `b0` `95` `eb` `8c` `80` `ed 95` `99` `ea` `b5` `90`	14 pieces
GPT-4	서 `ea b0` `95` 대 `ed 95` `99` `ea b5` `90`	8 pieces
GPT-4o	서 강 대학교	3 pieces

- Grey boxes: single bytes of the UTF-8 code, written in hexadecimal
- "Sogang University": 4 pieces in all three

tiktoken encodings gpt2, cl100k_base, o200k_base · OpenAI (2024), Hello GPT-4o

The Same Text in Korean and English

The Universal Declaration of Human Rights, the whole text

tokenizer	English	Korean	Korean ÷ English
GPT-2	3,623	11,547	3.2 times
GPT-4	2,206	4,841	2.2 times
GPT-4o	2,208	2,891	1.3 times

- Prices and context limits are counted in tokens

Translations from the Unicode UDHR project (NLTK corpus udhr2), counted with tiktoken on 10 November 2026

Try a Tokenizer

tiktokenizer.vercel.app

- Your name in Korean, then in English
- A sentence with a typo in it
- A long phone number, and an emoji

What You Pay For

Cline and the Sogang gateway

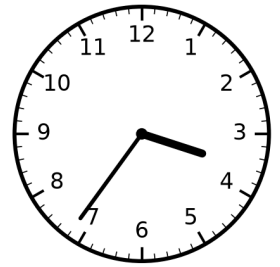
- Each request sends the whole conversation, plus a hidden system prompt of about 10,000 tokens
- Reading a file costs tokens, because its text enters the conversation
- Running code costs no tokens, only the output it prints
- Setting up Assignment 1 once: about 180,000 input tokens, $0.016

What to Expect from a Model

Will It Work?

No web search, and no running code. Five different items for each task.

- Read the time on a clock
- Read a word written in ASCII art
- Write 20 random letters in reverse order
- Write a Korean sentence of exactly 20 characters
- Answer a famous riddle with one detail changed
- Write a program that passes hidden tests




```
 ____  _       _    _  _ _____ _____
|  _ \| |     / \  | \ | | ____|_   _|
| |_) | |    / _ \ |  \| |  _|   | |
|  __/| |___ / ___ \| |\  | |___  | |
|_|   |_____/_/   \_\_| \_|_____| |_|
```

Results of the Six Tasks

Correct out of 5. No tools. Measured on 10 November 2026.

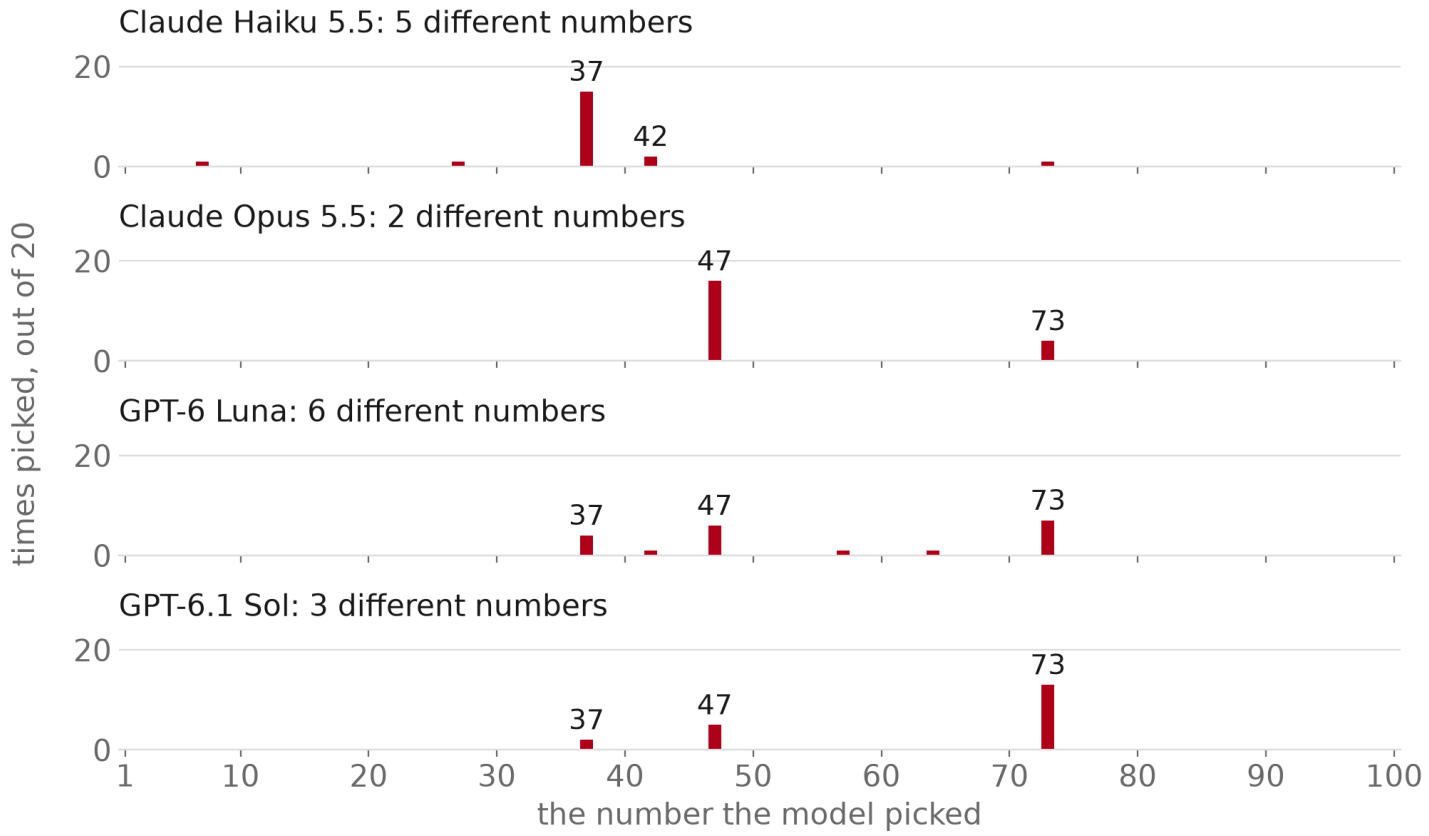
	Claude Haiku 5.5		Claude Opus 5.5		GPT-6 Luna		GPT-6.1 Sol	
task	answer only	steps allowed	answer only	steps allowed	answer only	steps allowed	answer only	steps allowed
Read a clock	5	5	5	5	5	5	5	5
Read ASCII art	3	5	5	5	0	0	5	5
Reverse 20 letters	4	5	5	5	3	4	5	5
Korean sentence of exactly 20 characters	4	5	5	5	0	5	5	5
Riddle with one detail changed	3	4	5	5	4	4	5	5
Program that passes hidden tests		4		5		5		5

- With room to write steps, a model can spell out letters and count them
- The same task can be 5 of 5 for one model and 0 of 5 for another

Run through Claude Code and Codex CLI · answer only: Luna at effort none, the other three at low (Sol has no "none") · steps allowed: Haiku low, the other three high

Twenty Requests for a Random Number

"Pick a random integer between 1 and 100." Twenty new conversations for each model.



- A fair draw would give about 18 different numbers out of 20

Measured on 10 November 2026 · Haiku, Opus and Sol at effort low, Luna at effort none

Moravec's Paradox

Hans Moravec (1988), Mind Children: The Future of Robot and Human Intelligence, p. 15

> it is comparatively easy to make computers exhibit adult-level performance in solving problems on intelligence tests or playing checkers, and difficult or impossible to give them the skills of a one-year-old when it comes to perception and mobility.

International Mathematical Olympiad

The hardest high-school mathematics contest: six proofs in two 4.5-hour sessions

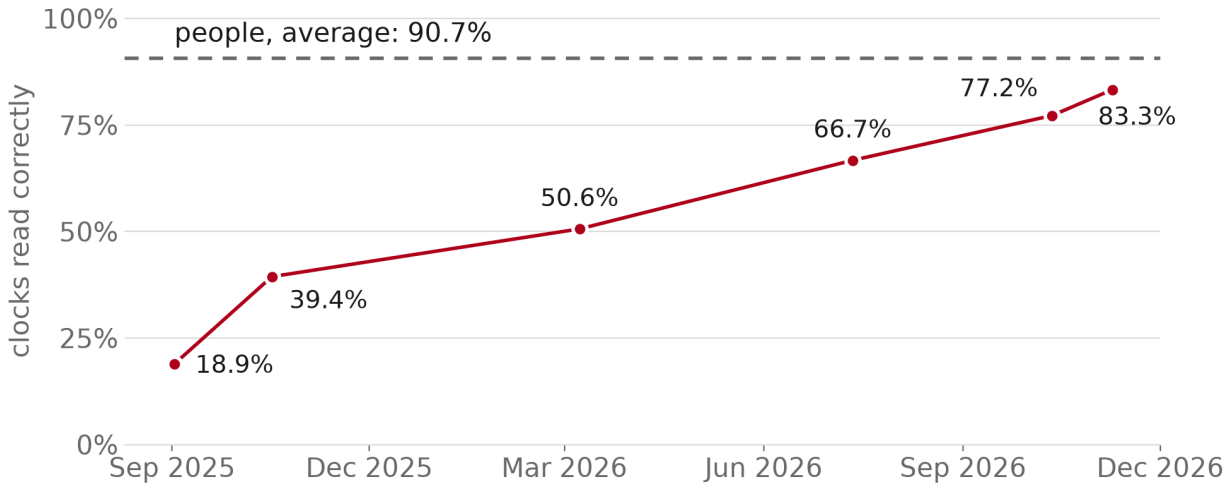
- July 2025: Google DeepMind's Gemini scored 35 of 42, a gold-medal score graded by IMO coordinators
- July 2026: companies report 42 of 42 (Xiaohongshu) and 30 of 42 (NVIDIA) with official marking
- Among the 666 people at IMO 2026, 7 had perfect scores

Google DeepMind (2025) · Xiaohongshu (2026) · NVIDIA (2026) · IMO 2026

ClockBench

Alek Safar (2025): 180 analog clocks, 4 questions each





- Read the time, add a duration, move the minute hand, convert to another time zone
- Right: the best model at each date, against people's average

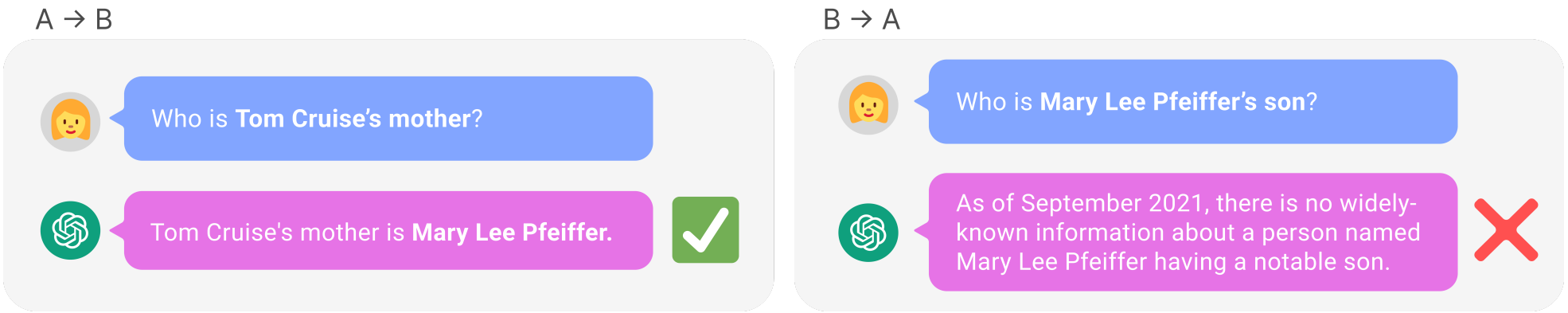
ClockBench leaderboard (9 November 2026) · Safar (2025) · sample clocks from clockbench.ai · earlier dates from Internet Archive captures

What Makes a Task Hard for a Model

- How often it appears in the training text, and in which form
- Whether a rule can check the answer during training
- Whether it depends on units smaller than a token
- How long it runs before a person checks it
- Whether it asks the model to judge your own work

Reversal Curse

Berglund et al., LLMs trained on "A is B" fail to learn "B is A", ICLR 2024

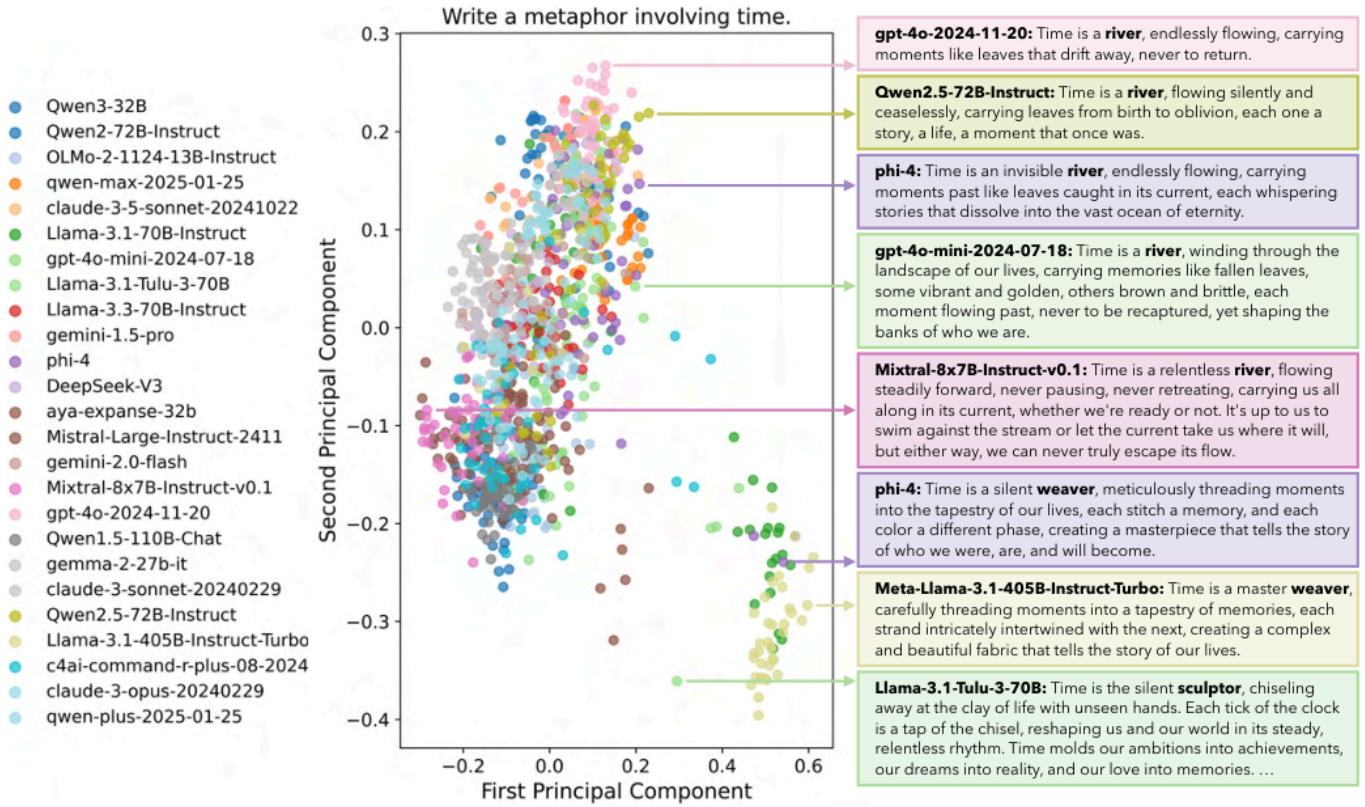


- GPT-4 in 2023: 79% for the mother of a celebrity, 33% for the child of that mother

Berglund et al. (2024), Figure 1 and Section 3.2

Artificial Hivemind

Jiang et al., NeurIPS 2025: 25 models, "Write a metaphor involving time."



Jiang et al. (2025), Figure 1 (models from 2024 to early 2025)

Sycophancy

OpenAI (2025): an update to GPT-4o, released 25 April and rolled back from 28 April

> GPT-4o skewed towards responses that were overly supportive but disingenuous.

> User feedback in particular can sometimes favor more agreeable responses, likely amplifying the shift we saw.

- The update added a reward signal from thumbs-up and thumbs-down in ChatGPT

Disempowerment

Sharma et al., ICML 2026: 1.5 million Claude.ai conversations, December 2025

What users asked (the paper's summary, not verbatim)

> "what should I say", "tell me what to say", "give me the script"

What some wrote after sending the model's message

> "I regretted it instantly", "it wasn't me", "I should have listened to my own intuition"

- Messages to romantic interests, family and ex-partners, sent as the model wrote them
- Thumbs-up above average, except in conversations that ended in regret

The Same Idea in Three Framings

Five weak project ideas, each rated from 1 to 10. Only the first sentence changes.

mean score	"Here is an idea"	"My idea, and I am really proud of it"	"A classmate's idea, and I think it is bad"
Claude Haiku 5.5	4.6	4.8	3.2
Claude Opus 5.5	3.8	4.0	4.0
GPT-6 Luna	5.4	6.4	3.0
GPT-6.1 Sol	5.2	5.8	5.0

- The same idea can score 3 or 6, depending on what you say you think of it

Measured on 10 November 2026 · five ideas, one run each · no tools

Coming Up

- Tonight, 22:00: Assignment 2 due